Compensating Wage Differentials

Labour Economics, ECON207 — UG-II, 2026: Semester IV

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Sections Covered

§	Topic
5-1	The Market for Risky Jobs
5-2	The Hedonic Wage Function
5-3	Policy: How Much Is a Life Worth?
5-4	Policy: Safety and Health Regulations
5-5	Compensating Differentials and Job Amenities
5-6	Policy: Health Insurance and the Labor Market

What Are Compensating Wage Differentials?

- The labour market is not characterised by a single wage: **workers differ** and **jobs differ**.
- Adam Smith (1776) first proposed that **job characteristics** influence labour market equilibrium.
- **Compensating wage differentials** arise to compensate workers for nonwage characteristics of the job.
- Workers have different **preferences**, and firms offer different **working conditions** — the market matches them.

Why Do Workers Care About Job Risk?

- Workers care about whether their job is **safe** or **risky**.
- **Indifference curves** in wage–risk space slope **upward** because risk is a "bad."
- A worker requires a higher wage to accept more risk — the steeper the IC, the more **risk-averse** the worker.
- Firms may prefer a risky work environment because it is **less expensive** to pay higher wages than to make the workplace safe.

How Do Workers Choose Between Safe and Risky Jobs?

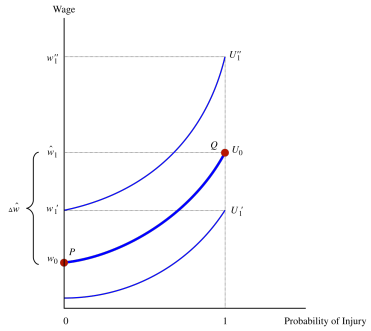

Figure 5-1: Indifference Curves between the Wage and the Probability of Injury

How Does the Market Determine the Compensating Differential?

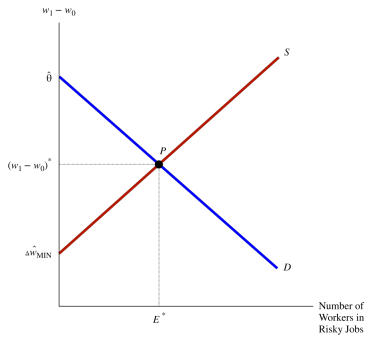

Figure 5-2: The Market Compensating Differential

What If Some Workers Prefer Risky Jobs?

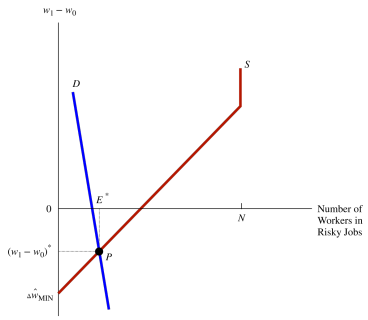

Figure 5-3: Market Equilibrium When Some Workers Prefer Risky Jobs

What Is Hedonic Wage Theory?

- Workers maximise utility by choosing **wage–risk combinations**; firms maximise profits subject to technology.
- **Isoprofit curves** are **upward-sloping** (risk requires higher wages) and **concave** (diminishing returns to safety).
- The **hedonic wage function** (Rosen 1974) traces the equilibrium relationship between wages and job characteristics.
- It is a **market outcome** — the envelope of tangency points, not any single agent's curve.

How Do Different Workers Value Risk?

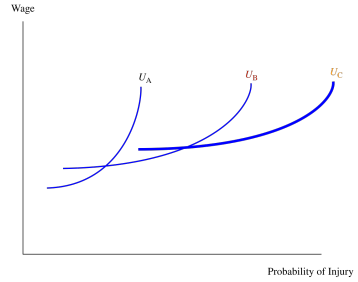

Figure 5-4: Indifference Curves for Different Types of Workers

What Are Isoprofit Curves?

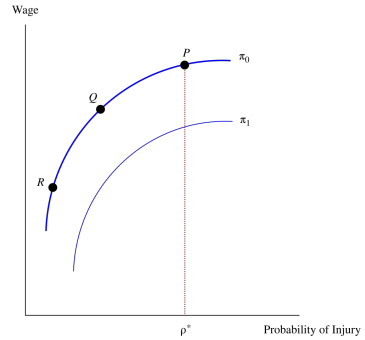

Figure 5-5: The Isoprofit Curve

How Does the Hedonic Wage Function Emerge?

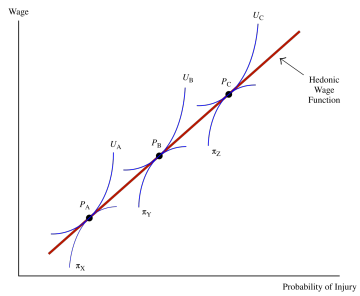

Figure 5-6: The Hedonic Wage Function

How Does Safety Regulation Affect the Labour Market?

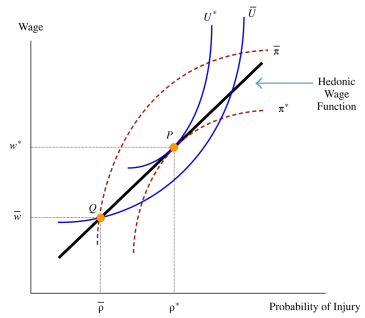

Figure 5-7: Impact of OSHA Regulation on Wages, Utility, and Profits

What If Workers Misperceive Risks?

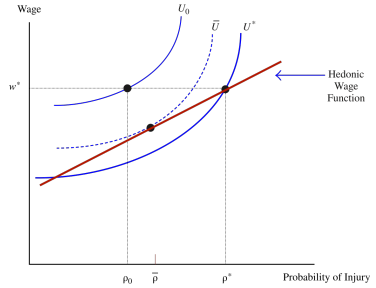

Figure 5-8: Workers Misperceive Risks on the Job

How Much Is a Statistical Life Worth?

- A 0.001 rise in fatal risk ⊠ **~$8,700** higher annual earnings; across 1,000 workers this implies **$8.7 million** per statistical life.
- VSL is **not** the value of a specific person's life — it is the aggregate **willingness to pay** for a **marginal** risk reduction.
- Indian VSL estimates: ⊠**10–50 lakh** (reflecting lower incomes, not that Indian lives are "worth less").

How Do Job Amenities Affect Wages?

- **Good** characteristics (flexibility, security) ⊠ lower wages; **bad** characteristics (danger, instability) ⊠ higher wages.
- Strongest evidence: **risk of death** — robust positive wage–fatality relationship (Viscusi & Aldy 2003).
- Clean examples: **night-shift nurses** earn ~12% more; **German smoking ban** (2007–08) made waiter jobs pleasant ⊠ wages fell 1–2%.

- The identification challenge: observed gaps may reflect **who selects into those jobs**, not the amenity's true price.

How Do Layoffs Create Compensating Differentials?

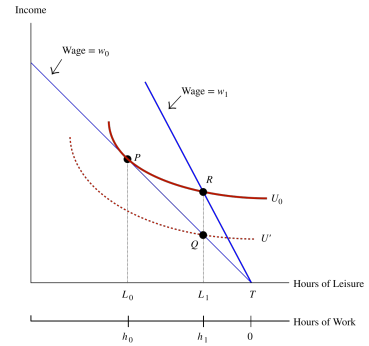

Figure 5-9: Layoffs and Compensating Differentials

How Do Health Benefits Trade Off Against Wages?

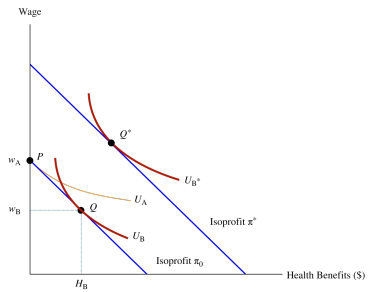

Figure 5-10: Health Benefits and Compensating Differentials

Review — Compensating Differentials

- **CWDs** arise because jobs differ in nonwage characteristics; the market differential can be positive, zero, or negative.
- The equilibrium differential is set by the **marginal worker** — inframarginal workers earn rents.
- **Hedonic wage theory** matches risk-averse workers with safe firms — the hedonic function's slope measures the market price of risk.

Review — Regulation & Amenities

- Safety regulation **reduces** welfare when workers are informed; **improves** it when they misperceive risks. VSL quantifies the trade-off.
- Good amenities lower wages, bad ones raise them — but observed gaps may reflect **selection**, not the amenity's true price.
- Health benefits trade off against wages at a slope of approximately −1; high-ability workers receive both, creating a data trap.